

# Repsol Electra 2X

Paraffinic dielectric oil — Inhibited

## PRODUCT DATA SHEET

### Description

ELECTRA 2X is a light, paraffinic, inhibited oil designed for use as insulating fluid in electrical equipment such as power and distribution transformers, circuit breakers, rheostats, etc. This oil has a very low freezing point, making it ideal for cold climates. It is manufactured from highly-refined treated base oils ensuring the absence of solid, polar or water compounds and of corrosive sulphur.

### Classification and standards

ELECTRA 2X is an inhibited oil formulated with antioxidant additives. It meets the IEC 60296 and ASTM D3487 Type II standards and has a very low freezing point, making it ideal for cold climates.

### Technical specifications

Values according to IEC 60296: 2012 Inhibited.

| Properties                                               | Units              | Method      | Min. | Max.                   | Typical value |
|----------------------------------------------------------|--------------------|-------------|------|------------------------|---------------|
| <b>Physical properties</b>                               |                    |             |      |                        |               |
| Appearance                                               |                    | Visual      |      | Shiny and transparent  |               |
| Density at 20 °C                                         | g/cm <sup>3</sup>  | ISO 12185   |      | 0.895                  | 0.82          |
| Viscosity at 100 °C                                      | mm <sup>2</sup> /s | ASTM D445   |      | 3.0                    |               |
| Viscosity at 40 °C                                       | mm <sup>2</sup> /s | ASTM D445   |      | 12                     | 11.1          |
| Viscosity at -30 °C                                      | mm <sup>2</sup> /s | ASTM D445   |      | 1,800                  | 477           |
| Freezing point                                           | °C                 | ISO 3016    |      | -40                    | -65           |
| Interfacial tension                                      | mN/m               | UNE 21320   | 40   |                        | 52            |
| <b>Chemical properties</b>                               |                    |             |      |                        |               |
| Acidity                                                  | mg KOH/g           | IEC 62021-2 |      | 0.01                   | <0.01         |
| Water content                                            | mg/kg              | IEC 60814   |      | 30                     | 11.5          |
| Corrosive sulphur                                        |                    | ASTM D1275B |      | Non-corrosive          |               |
| Inhibitor content                                        | % weight           | IEC 60666   |      | 0.4                    | 0.3           |
| DBDS                                                     | mg/kg              | IEC 62697-1 |      | Not detectable         |               |
| Furfural content                                         | mg/kg              | IEC 61198   |      | Not detectable (<0.05) |               |
| <b>Electrical properties</b>                             |                    |             |      |                        |               |
| Dielectric loss factor at 90 °C                          |                    | IEC 60247   |      | 0.005                  | 0.0006        |
| Breakdown voltage, treated                               | kV                 | IEC 60156   | 30   |                        | 60            |
| Breakdown voltage, untreated                             | kV                 | IEC 60156   | 70   |                        | >70           |
| <b>Oxidation stability — IEC 61125C (500 h / 120 °C)</b> |                    |             |      |                        |               |

| Properties                                | Units    | Method     | Min.   | Max. | Typical value |
|-------------------------------------------|----------|------------|--------|------|---------------|
| Total acidity                             | mg KOH/g | IEC 61125C |        | 0.3  | 0.179         |
| Sludge                                    | % weight | IEC 61125C |        | 0.05 | <0.01         |
| Dielectric loss factor at 90 °C           |          | IEC 61125C |        | 0.05 | 0.0010        |
| <b>Health, safety and the environment</b> |          |            |        |      |               |
| Flash point                               | °C       | ASTM D93   | 135    |      | 193           |
| PCA content                               | % weight | IP 346     |        | 3    | 0.8           |
| PCB content                               | % weight | IEC 61619  | Exempt |      | Not detected  |

*Note: Typical values correspond to current production. Whilst future production will conform to Repsol's specification, variations in these characteristics may occur. The guaranteed minimum/maximum values define the product specification.*